Achieving Envy-Freeness with Limited Subsidies under Negative Dichotomous Valuations

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Abstract

We study the problem of allocating indivisible *chores* among agents in a fair manner, with subsidies. Envy-free allocations of indivisible items are not guaranteed to exist, but envy-freeness can be achieved by additionally providing some subsidy to the agents; in the chores setting this subsidy is naturally read as compensation paid to the agents who absorb the workload. We study the subsidy required under *negative dichotomous* valuations, i.e., among agents whose marginal disutility for any chore is either zero or one: $v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\}$ for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$. This is the exact mirror of the dichotomous goods model of Barman et al. [BKNS22], who prove that there a per-agent subsidy of 0 or 1 always suffices, and we conjecture that the same guarantee holds for chores.

The mirror is not a relabelling. Subsidies are one-signed — money is always a good — so negating an instance negates the valuations but not the money, and the goods theory does not transfer by symmetry. This report sets out what does transfer and what the chore problem looks like once it has. The Halpern–Shah characterisation of envy-freeable allocations [HS19] survives verbatim, since it assumes nothing about the sign or the structure of the valuations, and it delivers an *integral* minimum subsidy vector here; the matching lower bound of $n - 1$ survives as well. Existing work already gives a bounded subsidy: negative dichotomous valuations are doubly monotone under the two-sided normalisation, so the reduction of Kawase et al. [KMS+24] yields $n - 1$ per agent and $n(n - 1)/2$ in total. The open question is whether the far stronger $\{0, 1\}$ guarantee of [BKNS22] survives the passage from goods to chores, a factor of n below that baseline.

We then isolate three structural facts about the chore problem: an arc-update lemma reducing the entire goods/chore asymmetry to two lines; a two-tier characterisation of the allocations admitting a $\{0, 1\}$ subsidy; and a size-shift transform exhibiting the chore problem as *exactly* the goods problem of [BKNS22] together with a cardinality-balancing requirement, which is the precise sense in which the conjecture does not follow from that work as a black box.

1 Introduction

Discrete fair division studies the allocation of resources that cannot be fractionally assigned among agents with individual preferences, and sits at the interface of mathematical economics and computer science [BCE+16, End17]. The classical fairness criterion is *envy-freeness* [Fol67, Var74]: no agent should prefer the bundle assigned to another agent over her own. For indivisible items an envy-free allocation need not exist, as the standard one-item two-agent instance shows, and a large body of work therefore studies relaxations such as envy-freeness up to one good (EF1) [Bud11, LMMS04].

A second and complementary line of work retains *exact* envy-freeness and buys off the residual envy with money. Each agent i receives, in addition to her bundle, a subsidy $p_i \geq 0$ supplied by a third party, and one asks for an allocation together with a subsidy vector under which every agent weakly prefers her own bundle-plus-subsidy to that of every other agent. The question is how much money this requires. Maskin [Mas87] answered it for unit-demand agents with $m = n$: under the scaling in which no agent values any single item above 1, a total subsidy of $n - 1$ suffices. Halpern and Shah [HS19] moved to the multi-demand setting with arbitrary m, gave the characterisation of *envy-freeable* allocations that the whole line now runs on, showed that a subsidy of $(n - 1)m$ is necessary and sufficient in the worst case when the allocation is handed to us, and conjectured that $n - 1$ suffices when we may choose the allocation. Brustle et al. [BDN+20] proved that conjecture in the stronger per-agent form — one dollar to each agent suffices for additive valuations, with the allocation simultaneously EF1 and balanced — and showed that for general monotone valuations $2(n - 1)$ per agent suffices, so that the total subsidy is $O(n^2)$ and, remarkably, *independent of the number of items*. Barman et al. [BKNS22] then improved the monotone bound by a factor of n on the dichotomous subclass: when every marginal $v_i(S \cup \{g\}) - v_i(S)$ lies in $\{0, 1\}$, there is an allocation whose accompanying subsidy is exactly 0 or 1 for each agent, hence at most $n - 1$ in total, computable in polynomial time in the value-oracle model. Their bound assumes neither additivity nor submodularity nor even subadditivity, and it is tight: with n agents and a single unit-valued good, $n - 1$ agents must each be paid 1.

Chores. Every result above is about *goods*. This paper asks the mirrored question, in which every item is a *chore*: an object that an agent would rather not hold, so that acquiring it weakly decreases her utility. Chores are not an exotic variant. Shift rotations, on-call duty, teaching and refereeing load, committee service, maintenance tasks and the liabilities attached to an estate are all allocation problems in which the items are burdens rather than prizes, and in all of them money is already the standard instrument of repair — as overtime pay, hardship allowance, or a course-release budget. If anything the subsidy model is more natural for chores than for goods, since compensating somebody for taking on unwanted work needs less justification than paying somebody for the privilege of not receiving a prize.

Concretely, we study *negative dichotomous* valuations: for every agent i, every $S \subseteq M$ and every $g \in M \setminus S$,

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\}.$$

Equivalently, each marginal $v_i(S \cup \{g\}) - v_i(S)$ is 0 or -1: every item is a chore for every agent, and taking on one more chore costs an agent either nothing or exactly one unit. This is the exact mirror of the model of [BKNS22], and we impose no further structure — in particular the valuations need not be additive, submodular, or subadditive.

Binary marginals are the natural model whenever a task either lands on an agent as a fresh unit of burden or is absorbed by a commitment she has already made. An engineer already staffing an on-call window absorbs a second incident in that window at no extra cost; a referee already reading for a venue absorbs one more paper from the same batch; a driver already routed through a district absorbs an extra stop along the way. Whether a given chore is free depends in an arbitrary way on the rest of the agent's assignment, which is exactly why the model must tolerate non-additive and even non-subadditive valuations, and it is the counterpart of the scheduling example used by [BKNS22] to motivate dichotomous valuations for goods.

Why this is not a change of sign. It is tempting to expect that a chores result follows from the corresponding goods result by negating the valuations. It does not, for three reasons that we record here because they shape everything that follows.

First, money is one-signed. The subsidy vector is constrained to $p \in \mathbb{R}^n_+$, and negating an instance negates the valuations but leaves the subsidies where they are. The polyhedron of feasible subsidy vectors is therefore a genuinely different object in the two models, and no formal duality carries a bound across.

Second, the obvious transformation is available only for two agents. Writing $c_i := -v_i$ for the cost form of a negative dichotomous valuation, the complement $\tilde{v}_i(S) := c_i(M) - c_i(M \setminus S)$ is again dichotomous in the sense of [BKNS22], so the class is closed under complementation. But the complement transformation respects the allocation only when $M \setminus A_j = A_i$, that is, only when $n = 2$. For $n \geq 3$ there is no such reduction, and that is where the content of the problem lies.

Third, and most tellingly, envy flows the other way. Under goods, envy concentrates *into* the agent holding the valuable items: many agents envy one, and the tight instance of [BKNS22] pays $n - 1$ agents one dollar each so that they stop envying the single agent who received the good. Under chores, envy emanates *out of* the agent carrying the load: one agent envies many. Since the minimum subsidy to an agent is the weight of the heaviest directed path leaving her in the envy graph [HS19, Theorem 2], the two situations are not interchangeable, and one should not assume in advance that the same bound is the right target.

What is already known, and what is open. Two facts delimit the problem before any work is done, and we state them explicitly so that the contribution can be measured against them.

On the one hand, existence of a bounded subsidy is not in question. Negative dichotomous valuations are doubly monotone — every item is a chore for every agent, which is a special case of every item being a good or a chore for each agent — and they satisfy the normalisation $|v_i(S \cup \{e\}) - v_i(S)| \leq 1$ that [KMS+24, §2] imposes. The reduction of Kawase et al., which converts any EF1 allocation into an envy-free one, therefore applies off the shelf and yields a subsidy of at most $n - 1$ per agent and $n(n - 1)/2$ in total [KMS+24, Theorem 1]. Two details make the containment work and are easy to get wrong. Their normalisation is two-sided, so it bounds chore disutility as well as value; and the EF1 they require is the two-sided form, which removes one item from *either* bundle, not the goods-only form of [HS19] — the two differ precisely when chores are present. An EF1 allocation is guaranteed to exist and to be computable in polynomial time for doubly monotone valuations, so the reduction has an input; for that step [KMS+24, §2] cites [LMMS04] and [BSV21], and the citation matters, because the envy-cycle argument of [LMMS04] does *not* port to chores unaltered and it is [BSV21] that supplies the repair.

On the other hand, the lower bound of the goods model survives intact.

Example 1 (The $n-1$ lower bound transfers)**.** Take n agents and $m = n-1$ chores, with $c_i(S) = |S|$ for every agent i — identical, binary, additive costs, which are in particular negative dichotomous. Every complete allocation leaves at least one agent empty-handed. Under the balanced allocation, in which $n-1$ agents hold one chore each and one agent holds nothing, each of the $n-1$ loaded agents envies the empty agent by exactly 1 and no other envy is present, so the minimum subsidy assigns 1 to each loaded agent and 0 to the empty one, for a total of $n-1$. No allocation does better: any complete allocation gives some agent a bundle of size $k \geq 1$ while some other agent is empty, and that agent's heaviest outgoing path already has weight k.

So the target statement is precisely the analogue of [BKNS22, Theorem 4].

Conjecture 2. *For every instance with negative dichotomous valuations there is an envy-free solution $(\mathcal{A}, p)$ with $p \in \{0,1\}^n$, hence with total subsidy at most $n-1$; and it can be computed in polynomial time given value-oracle access.*

Example 1 shows Conjecture 2 would be tight. Between the $n-1$ per agent that [KMS+24] already gives and the $\{0,1\}$ per agent being asked for lies a factor of n in the total, exactly the gap that [BKNS22] closed on the goods side. Our objective is to close it on the chores side or to exhibit an instance showing that it cannot be closed.

Scope of this report. Conjecture 2 is open. This report sets up the problem and stops there: Section 2 fixes the model and notation and records what transfers from the goods theory unchanged, and Section 3 isolates three structural facts about the chore problem — an arc-update lemma (Lemma 11) reducing the entire goods/chore asymmetry to two lines, a two-tier characterisation of $\{0,1\}$ solutions (Proposition 12), and a size-shift transform (Theorem 13) exhibiting the chore problem as exactly the goods problem of [BKNS22] together with a cardinality-balancing requirement. The last of these is the precise sense in which Conjecture 2 does not follow from [BKNS22] as a black box.

Conjecture 2 remains open. What stands in the way is recorded below: three techniques that settle the neighbouring corners of the square above, each of which fails here, and for each an explicit instance exhibiting the failure. Several special cases are settled; work towards the general statement is in progress and is not reported here.

Related Work. The subsidy line for goods is described above. Within it, the dichotomous subclass has attracted separate attention: Halpern and Shah [HS19] settle binary additive valuations with a total subsidy of $n-1$, Goko et al. [GIK+21] obtain the analogous bound for binary submodular valuations together with a strategy-proof mechanism, and Barman et al. [BKNS22] subsume both by dropping every structural assumption beyond binary marginals. Dichotomous preferences are themselves a well-studied restriction in social choice [BMS05, BEF21]. On the computational side, Caragiannis and Ioannidis [CI21] show that minimising the subsidy is NP-hard and give a fully polynomial-time approximation scheme for a constant number of agents, and Narayan et al. [NSV21] study the welfare cost of achieving envy-freeness with transfers.

The paper closest to the present setting is that of Kawase et al. [KMS+24], the only work in the envy-freeness-with-subsidy line that admits chores at all. Working with doubly monotone valuations under the two-sided normalisation, it decouples the two jobs that [BDN+20] performed together: rather than constructing a specific allocation and showing that it is cheap to subsidise, it takes *any* EF1 allocation as input and bounds the subsidy needed to convert it, obtaining $n-1$ per agent and $n(n-1)/2$ in total. Since EF1 allocations are known to exist for doubly monotone valuations, the wider class comes along at no cost. That reduction is what makes the present question well-posed rather than open-ended, and it is the baseline we are trying to beat.

The complementary repair — no money, weaker fairness notion — is pursued for binary valuations by Bu et al. [BSY23], who show that EFX allocations always exist and can be computed in polynomial time for general binary valuations, thus extending the matroid-rank result of Babaioff et al. [BEF21]. Under binary valuations one therefore has a clean dichotomy on the goods side: either pay 0 or 1 per agent and obtain exact envy-freeness [BKNS22], or pay nothing and obtain EFX [BSY23]. Whether either branch survives the passage to chores is open; this paper addresses the first.

The chores side. Work on indivisible chores has largely developed alongside rather than inside the subsidy line, and the two have met less often than one might expect. Where money has been brought to chores, it has been in the service of *proportionality* rather than envy-freeness: the question studied is the total subsidy needed to bring every agent down to her proportional share of the burden, for which the best bounds under additive disutilities normalised to at most 1 per chore are $n/4$ and then $n/3 - 1/6$, recently obtained together with Pareto optimality by Garg et al. [GSW25]. That is a different fairness notion and a different accounting; it does not bound the subsidy needed for envy-freeness.

The no-money branch, by contrast, has a direct chore analogue of [BSY23]. Tao et al. [TWYZ23] study chores with binary marginals and show that for binary *additive* costs an allocation that is simultaneously EFX and Pareto optimal always exists and is computable in polynomial time. So on the chores side the second half of the binary dichotomy — pay nothing, obtain EFX — is established at least for additive costs, while the first half, pay 0 or 1 per agent and obtain exact envy-freeness, is what Conjecture 2 asks for. That money is genuinely needed here is not obvious a priori but is known: Bhaskar et al. [BSV21] show that deciding whether an exactly envy-free allocation of chores exists is NP-complete already for binary additive costs, and give a polynomial-time EF1 algorithm for chores and for doubly monotone instances.

Where the two lines have met is on the *additive* side, and recently. Lu et al. [LMS26] prove the goods-and-chores analogue of [BDN+20]: for additive utilities in which each item is a good for some agents and a chore for others, with every $|v_i(g)| \le 1$, a subsidy of one dollar per agent suffices, the bound is tight, and the allocation is computable in polynomial time — hence $n-1$ in total. This settles the additive chore problem completely.

It does not settle ours, and the reason is exactly the reason [BKNS22] was not a corollary of [BDN+20]. Dichotomous valuations are not additive and additive valuations are not dichotomous; the two classes are incomparable. The four corners are

	goods	chores
additive	[BDN+20]	[LMS26]
dichotomous	[BKNS22]	**Conjecture 2**

each of the three settled corners giving one dollar per agent and $n-1$ in total. The present question is the fourth. Note also that [LMS26] does not displace [KMS+24] as the baseline of Section 1: for valuations that are dichotomous but not additive, $n-1$ per agent from [KMS+24] remains the best published bound.

We are not aware of any treatment of envy-freeness with subsidy for chores under dichotomous costs. This is the outcome of a literature search rather than a proof of novelty — [LMS26] appeared in July 2026, which is a reminder of how quickly the surrounding corners are being filled in — and it should be repeated before the paper is circulated.

2 Notation and Preliminaries

We study the allocation of m indivisible items among n agents, with subsidies. Write $N = [n]$ for the set of agents and M for the set of items, $|M| = m$. Each agent $i \in N$ has a valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$, and we represent an instance by the tuple $\langle N, M, \{v_i\}_{i\in N}\rangle$. Algorithms operate in the standard *value-oracle* model: for each v_i we assume only an oracle that returns $v_i(S)$ for a queried set $S \subseteq M$, and running time is measured in n, m and the number of oracle calls.

An *allocation* $\mathcal{A} = (A_1, \ldots, A_n)$ is an ordered tuple of pairwise disjoint *bundles*, with A_i assigned to agent i. The allocation is *complete* if $\bigcup_{i\in N} A_i = M$ and *partial* otherwise. The ordering matters throughout: much of the theory below concerns permuting a fixed family of bundles among the agents, and for a permutation $\sigma \in \mathbb{S}_n$ we write $\mathcal{A}_\sigma := (A_{\sigma(1)}, \ldots, A_{\sigma(n)})$ for the allocation in which agent i receives $A_{\sigma(i)}$. The utilitarian welfare of $\mathcal{A}$ is $\mathrm{SW}(\mathcal{A}) = \sum_{i\in N} v_i(A_i)$.

Definition 3 (Envy-Freeness)**.** *An allocation $\mathcal{A} = (A_1, \ldots, A_n)$ is* envy-free *(EF) iff $v_i(A_i) \geq v_i(A_j)$ for all agents $i, j \in N$.*

Envy-free allocations of indivisible items need not exist, so we allow a third party to supplement the allocation with money. Agents have quasi-linear utilities: agent i receiving bundle A_i and subsidy p_i enjoys utility $v_i(A_i) + p_i$, with money entering linearly and at the same exchange rate for every agent.

Definition 4 (Envy-Free Solution)**.** *An allocation $\mathcal{A} = (A_1, \ldots, A_n)$ and a subsidy vector $p = (p_1, \ldots, p_n) \in \mathbb{R}^n_+$ constitute an* envy-free solution *$(\mathcal{A}, p)$ iff $v_i(A_i) + p_i \geq v_i(A_j) + p_j$ for all $i, j \in N$.*

An allocation $\mathcal{A}$ is *envy-freeable* if some $p \in \mathbb{R}^n_+$ makes $(\mathcal{A}, p)$ an envy-free solution; this is a property of the allocation alone. Following [BKNS22] we write $\mathrm{M}(p) := \{i \in N \mid p_i \geq p_j \text{ for all } j \in N\}$ for the set of maximally subsidised agents.

We will occasionally need the standard relaxation of envy-freeness. Note that two inequivalent forms of it appear in this literature: the goods-only form of [HS19], which removes one item from the envied bundle, and the two-sided form used by [KMS+24], which removes one item from *either* bundle. They coincide when all items are goods and differ once chores are present, and only the two-sided form supports the doubly monotone results. We adopt the two-sided form.

2.1 Negative Dichotomous Valuations

Definition 5 (Negative Dichotomous Valuation)**.** *A valuation $v_i : 2^M \to \mathbb{R}$ with $v_i(\emptyset) = 0$ is* negative dichotomous *iff*

$$v_i(S) - v_i(S \cup \{g\}) \in \{0, 1\} \qquad \text{for every } S \subseteq M \text{ and every } g \in M \setminus S,$$

equivalently iff every marginal $\Delta_i^+(S, g) := v_i(S \cup \{g\}) - v_i(S)$ lies in $\{-1, 0\}$.

Every item is thus a chore for every agent: v_i is non-increasing, so $v_i(S) \geq v_i(T)$ whenever $S \subseteq T$. We stress that nothing further is assumed — the valuations need not be additive, submodular, or subadditive — which is the generality at which [BKNS22] works on the goods side.

It is often more convenient to work with the non-negative *cost form*

$$c_i := -v_i, \qquad \text{so} \qquad c_i(\emptyset) = 0, \quad c_i(S) \leq c_i(T) \text{ for } S \subseteq T, \quad c_i(S \cup \{g\}) - c_i(S) \in \{0, 1\}.$$

Definition 6 (Dichotomous cost function)**.** *A set function $c : 2^M \to \mathbb{R}$ is* dichotomous *if $c(\emptyset) = 0$ and every marginal $c(S \cup \{g\}) - c(S)$ lies in $\{0, 1\}$.*

Monotonicity is not a separate hypothesis: marginals in $\{0, 1\}$ are non-negative, so a dichotomous c is automatically non-decreasing. We nonetheless list it above because it is the property most often used directly.

The correspondence $v_i \leftrightarrow c_i$ is a bijection between negative dichotomous valuations and dichotomous cost functions in the sense of [BKNS22]: the class we study is exactly the pointwise negation of the class they study. In cost form an envy-free solution is the requirement $c_i(A_i) - p_i \leq c_i(A_j) - p_j$ for all i, j, read as "my own burden net of my compensation is no worse than what I perceive of yours". We use whichever of v_i and c_i is clearer locally and always state which.

2.2 The Envy Graph and the Halpern–Shah Characterisation

For an allocation $\mathcal{A}$, the *envy graph* $G_{\mathcal{A}}$ is the complete weighted directed graph on vertex set N in which the arc (i, j) carries weight

$$w_{\mathcal{A}}(i, j) := v_i(A_j) - v_i(A_i) = c_i(A_i) - c_i(A_j),$$

the envy that agent i has towards agent j. The weight of a directed path P is $w_{\mathcal{A}}(P) = \sum_{(i,j)\in P} w_{\mathcal{A}}(i, j)$, and $\ell_{\mathcal{A}}(i)$ denotes the maximum weight of any path in $G_{\mathcal{A}}$ starting at i, the empty path of weight 0 included.

The two results below are the engine of the entire subsidy line. Both are stated by Halpern and Shah [HS19] for additive valuations, but their proofs use only the arc weights and permutations of the bundles — neither additivity nor monotonicity nor any sign condition on v_i enters — and [BKNS22] invokes them in exactly this generality for (non-additive) dichotomous valuations. They therefore apply verbatim to negative dichotomous valuations, and we use them without further comment.

Theorem 7 ([HS19])**.** *For any allocation $\mathcal{A} = (A_1, \dots, A_n)$, the following are equivalent.*

(i) *$\mathcal{A}$ is envy-freeable.*

(ii) *$\mathcal{A}$ maximises utilitarian welfare across all reassignments of its own bundles among the agents: for every permutation σ over N, $\sum_{i \in N} v_i(A_i) \geq \sum_{i \in N} v_i(A_{\sigma(i)})$.*

(iii) *The envy graph $G_{\mathcal{A}}$ has no positive-weight directed cycle.*

Theorem 8 ([HS19])**.** *For any envy-freeable allocation $\mathcal{A}$, the subsidy vector p^* given by $p^*_i := \ell_{\mathcal{A}}(i)$ realises an envy-free solution $(\mathcal{A}, p^*)$, and $p^*_i \leq p_i$ for every $i \in N$ and every p such that $(\mathcal{A}, p)$ is an envy-free solution. It can be computed in strongly polynomial time by running an all-pairs longest-path computation on $G_{\mathcal{A}}$.*

Condition (ii) of Theorem 7 has the standing consequence that an envy-freeable allocation can always be manufactured from an arbitrary one: fix the bundles and reassign them according to a maximum-weight perfect matching between agents and bundles. What is at issue is never the existence of an envy-freeable allocation, only the size of the subsidy that Theorem 8 then charges for it.

Specialising to our model gives the following, which we use throughout.

Observation 9 (Integrality)**.** *Let $\{v_i\}_{i \in N}$ be negative dichotomous. Then $v_i(S) \in \mathbb{Z}_{\leq 0}$ for every i and every $S \subseteq M$; every arc weight $w_{\mathcal{A}}(i, j)$ is an integer; and for every envy-freeable allocation $\mathcal{A}$ the minimum subsidy vector satisfies $p^* \in \mathbb{Z}^n_{\geq 0}$.*

Proof Fix i and $S = \{g_1, \dots, g_k\}$. Telescoping from $v_i(\emptyset) = 0$ along any ordering of S writes $v_i(S)$ as a sum of k marginals, each of which lies in $\{-1, 0\}$ by Definition 5; hence $v_i(S) \in \mathbb{Z}$ and $-k \leq v_i(S) \leq 0$. Consequently each arc weight $w_{\mathcal{A}}(i, j) = v_i(A_j) - v_i(A_i)$ is a difference of integers, and the weight of any directed path, being a finite sum of arc weights, is an integer. By Theorem 8, $p^*_i = \ell_{\mathcal{A}}(i)$ is the maximum such weight over paths starting at i, and it is non-negative because the empty path has weight 0. □

Observation 9 is what makes the target statement — a subsidy vector in $\{0, 1\}^n$ — a statement about a *bound* rather than about rounding: once p^* is known to be integral, showing $p^*_i \leq 1$ for every i is the same as showing $p^* \in \{0, 1\}^n$. The same observation underlies the corresponding step in [BKNS22], where the arc weights of a dichotomous instance are likewise integers.

A second normalisation costs nothing and removes a clause from the target.

Proposition 10 (Some agent is always unpaid)**.** *For every envy-freeable allocation $\mathcal{A}$ there is an agent h with $p^*_h = 0$. Consequently, if $p^* \in \{0, 1\}^n$ then $\sum_{i \in N} p^*_i \leq n - 1$.*

Proof By Theorem 7(iii) the envy graph $G_{\mathcal{A}}$ has no positive-weight cycle, so a walk of maximum weight may be taken simple and $\ell_{\mathcal{A}}$ is finite; it is non-negative because the empty path qualifies. Choose a path P attaining $\max_i \ell_{\mathcal{A}}(i)$ and let h be its final vertex. If $\ell_{\mathcal{A}}(h) > 0$ there is a positive-weight walk Q leaving h, and P followed by Q is a walk from P's start of weight $w_{\mathcal{A}}(P) + w_{\mathcal{A}}(Q) > w_{\mathcal{A}}(P)$, contradicting the maximality of P among walks. Hence $\ell_{\mathcal{A}}(h) = 0$. The second claim is immediate: at most $n - 1$ coordinates of p^* can equal 1. □

Proposition 10 justifies the word "hence" in Conjecture 2: the bound on the *total* is not an independent requirement but a consequence of the per-agent one. The conjecture is therefore a single-clause statement,

$$\exists \mathcal{A} \ : \quad \max_{i \in N} \ell_{\mathcal{A}}(i) \ \leq \ 1,$$

and it is this form that the rest of the report works with.

3 Structure of the Chore Problem

Throughout this section we work in cost form, so $w_{\mathcal{A}}(i,j) = c_i(A_i) - c_i(A_j)$.

3.1 The arc-update table, and the whole of the asymmetry

Every incremental algorithm in this literature adds one item to one bundle and then asks what happened to the envy graph. The answer is two lines long, and comparing those two lines against their goods counterparts accounts for the entire difference between the present problem and that of [BKNS22].

For an allocation $\mathcal{A}$, an agent x and an unallocated chore $g \notin \bigcup_i A_i$, write

$$Z^x \ := \ (A_1, \ldots, A_x \cup \{g\}, \ldots, A_n)$$

for the allocation obtained by giving g to x, and put

$$\delta_x \ := \ c_x(A_x \cup \{g\}) - c_x(A_x), \qquad \delta_{i,x} \ := \ c_i(A_x \cup \{g\}) - c_i(A_x),$$

both of which lie in $\{0, 1\}$. So δ_x is what the chore costs x herself, and $\delta_{i,x}$ is what it costs in i's opinion.

Lemma 11 (Arc update)**.** *For all $i, j \in N$,*

$$w_{Z^x}(i,j) = w_{\mathcal{A}}(i,j) \quad (i,j \neq x), \qquad w_{Z^x}(x,j) = w_{\mathcal{A}}(x,j) + \delta_x, \qquad w_{Z^x}(i,x) = w_{\mathcal{A}}(i,x) - \delta_{i,x}.$$

Proof Only A_x changes, so an arc between two agents other than x has both endpoints untouched. For an arc leaving x, $w_{Z^x}(x,j) = c_x(A_x \cup \{g\}) - c_x(A_j) = w_{\mathcal{A}}(x,j) + \delta_x$. For an arc entering x, $w_{Z^x}(i,x) = c_i(A_i) - c_i(A_x \cup \{g\}) = w_{\mathcal{A}}(i,x) - \delta_{i,x}$. □

The mirror statement for goods is obtained by replacing costs with values: there the arcs *out of* the receiving agent fall and the arcs *into* her rise. The consequences are opposite in a way that matters.

	goods [BKNS22]	chores (here)
arcs out of the receiver	fall	**rise by** δ_x
arcs into the receiver	**rise**	fall by $\delta_{i,x}$
so the receiver must be	*maximally* subsidised	*minimally* subsidised
which is guaranteed by	$\mathrm{M}(p) \neq \emptyset$	$p_x = 0$ for some x

Both guarantees hold trivially — some agent is always maximally subsidised, and some agent always has $p_x = 0$ under the minimum subsidy vector. The asymmetry is subtler than the table suggests. Under goods, the harm done by an insertion is confined to paths *ending* at the receiver, and requiring $x \in \mathrm{M}(p)$ caps every such path at $p_u - p_x \le 0$ in one stroke. Under chores, the harm is done to paths *leaving* the receiver, and $p_x = 0$ caps only the paths that *start* at x. A path that merely passes through x — entering along an arc from some i with $\delta_{i,x} = 0$ and leaving along an arc that has just risen by $\delta_x = 1$ — is not controlled by any condition on p_x. Whether that gap can actually be realised is the question an incremental algorithm has to answer.

3.2 The shape of a $\{0,1\}$ solution

Proposition 12 (Two-tier characterisation). *Let $p \in \{0,1\}^n$ and put $S := \{i \in N \mid p_i = 1\}$. Then $(\mathcal{A}, p)$ is an envy-free solution if and only if*

(a) $c_i(A_i) \le c_i(A_j)$ whenever i, j lie both in S or both in $N \setminus S$;

(b) $c_i(A_i) \le c_i(A_j) + 1$ for $i \in S$ and $j \in N \setminus S$;

(c) $c_i(A_i) + 1 \le c_i(A_j)$ for $i \in N \setminus S$ and $j \in S$.

Proof In cost form Definition 4 reads $c_i(A_i) \le c_i(A_j) + p_i - p_j$ for all i, j. Substituting the three possible values $0, 1, -1$ of $p_i - p_j$ gives (a), (b) and (c) respectively. □

The reading is worth keeping in mind when designing an algorithm. The unpaid agents are exactly envy-free among themselves and *strictly* prefer their own bundle to that of every paid agent, by a full unit; the paid agents are exactly envy-free among themselves and envy-free up to one unit towards the unpaid. This is a certificate format, and it also suggests choosing the partition $S \mid N \setminus S$ *first* and allocating afterwards, rather than building the allocation up one chore at a time.

3.3 The size-shift transform

Complementation relates the chore and goods models only for $n = 2$. There is a second transform, available for every n, which does not solve the problem but says precisely how far it is from being a corollary of [BKNS22].

Theorem 13 (Size shift). *For a negative dichotomous instance with cost functions c_i, define $\tilde{v}_i(S) := |S| - c_i(S)$. Then each $\tilde{v}_i$ is a dichotomous* goods *valuation in the sense of [BKNS22], and for every allocation $\mathcal{A}$ and all i, j,*

$$w_{\mathcal{A}}(i,j) \;=\; \tilde{w}_{\mathcal{A}}(i,j) \;+\; |A_i| - |A_j|,$$

where $\tilde{w}_{\mathcal{A}}$ denotes the envy graph of the goods instance $\{\tilde{v}_i\}$. Consequently every cycle *has the same weight in both graphs — so $\mathcal{A}$ is envy-freeable for the chores $\{c_i\}$ if and only if it is envy-freeable for the goods $\{\tilde{v}_i\}$ — while for a path P from u to v,*

$$w_{\mathcal{A}}(P) \;=\; \tilde{w}_{\mathcal{A}}(P) + |A_u| - |A_v|, \qquad \textit{hence} \qquad \ell_{\mathcal{A}}(u) \;=\; \max_{v \in N}\left[\,\tilde{d}_{\mathcal{A}}(u,v) + |A_u| - |A_v|\,\right],$$

with $\tilde{d}_{\mathcal{A}}(u,v)$ the maximum weight of a path from u to v in the goods graph and $\tilde{d}_{\mathcal{A}}(u,u) = 0$.

Proof $\tilde{v}_i(\emptyset) = 0$, and for $g \notin S$, $\tilde{v}_i(S \cup \{g\}) - \tilde{v}_i(S) = 1 - (c_i(S \cup \{g\}) - c_i(S)) \in \{0, 1\}$, so $\tilde{v}_i$ is dichotomous. For the identity,

$$\tilde{w}_{\mathcal{A}}(i, j) = \tilde{v}_i(A_j) - \tilde{v}_i(A_i) = \left(|A_j| - c_i(A_j) \right) - \left(|A_i| - c_i(A_i) \right) = w_{\mathcal{A}}(i, j) - |A_i| + |A_j| .$$

Summing along a path $(i_1, \ldots, i_r)$, the terms $|A_{i_t}| - |A_{i_{t+1}}|$ telescope to $|A_{i_1}| - |A_{i_r}|$; along a cycle they telescope to 0. The characterisation of envy-freeability by the absence of positive cycles (Theorem 7) then transfers, and the formula for $\ell_{\mathcal{A}}(u)$ follows from Theorem 8. □

4 Obstructions to the Natural Approaches

Three techniques settle the neighbouring corners of the square in Section 1: build the allocation one item at a time, as in [BKNS22]; select a welfare-maximising allocation, as the exchange structure of dichotomous valuations invites; or reduce to a goods instance and price the residual constraint away. Each fails for negative dichotomous valuations, and each failure is witnessed by a single small instance. We treat the first two here; the third needs a construction of its own and is deferred to Section 5. These obstructions delimit what a proof of Conjecture 2 can look like, and two of the tools built along the way are used throughout.

4.1 Incremental insertion

The algorithm of [BKNS22] maintains an envy-free solution over a partial allocation with $p \in \{0, 1\}^n$, allocates one further item at each step, and repairs the invariant by permuting whole bundles. Nothing already allocated is ever moved. Two steps of that template come out *easier* on the chore side.

Theorem 14 (Free insertion)**.** *Let $\mathcal{A}$ be envy-freeable and let g be an unallocated chore with $c_x(A_x \cup \{g\}) = c_x(A_x)$ for some agent x. Then the allocation obtained by giving g to x is envy-freeable, and its subsidy vector is pointwise no larger than that of $\mathcal{A}$.*

Proof By Lemma 11 the arcs between agents other than x are unchanged; the arcs out of x are unchanged, the marginal being 0; and the arcs into x weakly decrease. Every arc weight therefore weakly decreases, so no positive-weight cycle can appear and no path weight can rise. □

For goods the corresponding step needs the receiver to be maximally subsidised, a permutation of the bundles, and an extendability argument, because there the insertion *raises* the arcs into the receiver. Here it lowers them. The only case requiring work is a chore whose marginal cost is 1 for every agent on her own bundle.

The second tool is the cheapest general handle on $\ell_{\mathcal{A}}$ available, and is used repeatedly below.

Theorem 15 (Cycle-closing bound)**.** *For any envy-freeable allocation $\mathcal{A}$ and any agent u,*

$$\ell_{\mathcal{A}}(u) \;\leq\; \max\Big(0,\, \max_{v \neq u} \left[\, c_v(A_u) - c_v(A_v) \,\right]\Big).$$

Proof Since $\mathcal{A}$ is envy-freeable, $G_{\mathcal{A}}$ has no positive-weight directed cycle (Theorem 7(iii)). Every closed walk decomposes into simple cycles and so also has weight at most 0; deleting a closed sub-walk therefore never decreases a walk's weight, and every walk starting at u is dominated by a *simple* path starting at u. As there are finitely many simple paths, the maximum defining $\ell_{\mathcal{A}}(u)$ is attained, by a simple path P^*.

If P^* is empty then $\ell_{\mathcal{A}}(u) = 0$ and the claim holds, the right-hand side being non-negative. Otherwise P^*, being simple, does not revisit u, so its terminal vertex v satisfies $v \neq u$ and appending the arc (v, u) yields a simple cycle C with $w_{\mathcal{A}}(C) \leq 0$. Hence

$$\ell_{\mathcal{A}}(u) \;=\; w_{\mathcal{A}}(P^*) \;=\; w_{\mathcal{A}}(C) - w_{\mathcal{A}}(v,u) \;\leq\; -w_{\mathcal{A}}(v,u) \;=\; c_v(A_u) - c_v(A_v),$$

which is at most the maximum over $v \neq u$. ◻

Corollary 16. *If $c_v(A_u) \leq c_v(A_v) + 1$ for all $u, v \in N$ — no agent would suffer more than one extra unit by taking anybody else's bundle — then $p^* \in \{0,1\}^n$.*

The bound is an *upper estimate*, not an evaluation. Here $\ell_{\mathcal{A}}(u)$ is the maximum weight of a path out of u, whereas the right-hand side retains only the endpoint of that path, and the inequality is in general strict: if $u \to a \to b$ has weight 2 and $w_{\mathcal{A}}(b, u) = -3$, the cycle weighs -1, so $\mathcal{A}$ is envy-freeable with $\ell_{\mathcal{A}}(u) = 2$ while the bound returns 3. Evaluating $\ell_{\mathcal{A}}$ requires a longest-path computation; the bound costs $O(n)$ arc lookups and is used only where an upper estimate suffices, as in Corollary 16.

The template nevertheless cannot be completed.

Example 17 (Insertion cannot be completed)**.** Take $M = \left\{a_1, a_2, g\right\}$ and

$$c_1(S) = \max\left(0, |S| - 1\right), \qquad c_2(S) = c_3(S) = |S|\,,$$

all dichotomous. The partial allocation $\mathcal{A} = (\{a_1, a_2\}, \emptyset, \emptyset)$ has envy matrix

$$\left(w_{\mathcal{A}}(i,j)\right) = \begin{pmatrix} 0 & 1 & 1 \\ -2 & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix},$$

with no positive cycle and $p^* = (1, 0, 0)$: a legal state of the invariant. The marginal cost of g is 1 for every agent on her own bundle, and inserting g into each bundle in turn gives minimum subsidy vectors $(2, 0, 0)$, $(2, 1, 0)$ and $(2, 0, 1)$ — each with an entry equal to 2. Yet the complete allocation $(\{a_1\}, \{a_2\}, \left\{g\right\})$ has an identically zero envy matrix and needs no subsidy at all.

Note that c_1 is supermodular, which is permitted: neither [BKNS22] nor the present work restricts to submodular valuations, and the instance genuinely needs it. The conclusion is that any proof of Conjecture 2 must be free to **re-open already-allocated bundles**; repairing by permutation alone is provably insufficient.

4.2 Selecting a utilitarian optimum

If the allocation cannot be built incrementally, it may instead be *selected* from a set small enough to reason about. Utilitarian optimality is the obvious candidate, for two good reasons. By Theorem 7 an allocation minimising $\sum_i c_i(A_i)$ over *all* allocations is automatically envy-freeable, so half of what is needed comes free. And it carries an exchange property: if $g \in A_i$ satisfies $c_i(A_i) - c_i(A_i \setminus \{g\}) = 1$, then $c_j(A_j \cup \{g\}) = c_j(A_j) + 1$ for every j, since otherwise the allocation would not have been optimal.

Example 18 (The utilitarian optimum can be strictly worse)**.** Take $M = \{g_1, g_2, g_3, g_4\}$ with the costs below; each column is 0 at $\emptyset$ with every marginal in $\{0, 1\}$, so all three are dichotomous.

S	c_1	c_2	c_3	S	c_1	c_2	c_3
$\emptyset$	0	0	0	g_3g_4	2	1	1
g_1	1	1	1	$g_1g_2g_3$	2	3	2
g_2	1	1	1	$g_1g_2g_4$	3	2	2
g_3	1	1	0	$g_1g_3g_4$	2	2	2
g_4	1	1	1	$g_2g_3g_4$	2	2	2
g_1g_2	2	2	2	$g_1g_2g_3g_4$	3	3	3
g_1g_3	1	2	1				
g_1g_4	2	2	2				
g_2g_3	1	2	1				
g_2g_4	2	2	2				

Agent 1 pays 1 for every singleton and agent 3 pays 0 only for $\{g_3\}$, so any allocation of total cost below 2 would force $A_2 \supseteq \{g_1, g_2, g_4\}$, costing agent 2 at least 2. The minimum total cost is therefore 2, attained *only* by $\mathcal{A}^\star = (\emptyset, \{g_1, g_2, g_4\}, \{g_3\})$, whose envy graph has $w_{\mathcal{A}^\star}(2, 1) = 2$ and hence $p^* = (0, 2, 0)$. But $\mathcal{B} = (\{g_1, g_3\}, \{g_2\}, \{g_4\})$, of total cost 3, has every arc at most 0 and needs no subsidy.

The optimum here is *unique*, so there is no tie to break: every rule selecting a nonempty subset of the utilitarian-optimal allocations returns $\mathcal{A}^\star$ and pays 2, while a subsidy-free allocation exists outside that set. The gap is the largest possible in one step, for one extra unit of total burden. Any search procedure must therefore be free to **give up total cost**.

5 The Replica Instance

The third of the techniques named in Section 4 turns the chore problem into a *goods* problem, to which the theorem of [BKNS22] applies verbatim, and then enforces what is left over by pricing. The translation is exact and is worth having in its own right; the pricing step is the third obstruction, and is treated in Section 5.2.

5.1 The construction

Fix an instance with cost functions c_i on chores $M = \{a_1, \ldots, a_m\}$. The *replica instance* $\hat{I}$ carries $n-1$ copies of a good b_j for each chore a_j,

$$\hat{M} := \left\{ b_j^{(1)}, \ldots, b_j^{(n-1)} \,:\, j \in M \right\},$$

and for $S \subseteq \hat{M}$, writing $\tau(S) \subseteq M$ for the set of *types* occurring in S,

$$\hat{v}_i(S) := c_i(M) - c_i\big(M \setminus \tau(S)\big).$$

The reading is that *handing agent i a copy of b_j relieves i of chore a_j*. Each chore ends with exactly one owner, so exactly $n-1$ agents must be relieved of it — hence $n-1$ copies — and no agent may be relieved of the same chore twice, so no bundle may hold two copies of one type. Call an allocation of $\hat{M}$ satisfying that last condition a *coverage allocation*. The condition is load-bearing rather than cosmetic: it is what makes $\hat{v}_i$ well defined as a dual, a second copy of a type having marginal 0 and never +1 again.

Theorem 19 (Replica transform)**.** *Let $c_1, \ldots, c_n$ be dichotomous. Then*

(i) each $\hat{v}_i$ is a dichotomous valuation on $\hat{M}$ in the sense of [BKNS22];

(ii) the map Φ sending an allocation $\mathcal{A}$ of M to the allocation in which bundle i receives one copy of each type in $M \setminus A_i$ is a bijection between partitions of M and coverage allocations of $\hat{M}$, up to relabelling copies of a type;

(iii) corresponding allocations have identical envy graphs: $\hat{w}_B(i,k) = w_{\mathcal{A}}(i,k)$ for all i, k.

Since the envy graphs agree arc for arc, so do the feasible subsidy vectors, and the $\{0,1\}$ target transfers in both directions without loss. The problem is therefore not merely analogous to a dichotomous goods problem; on coverage allocations it *is* one.

Corollary 20 (Coverage is the entire gap)**.** *Conjecture 2 holds if and only if every replica instance $\hat{I}$ admits a coverage allocation B with $\ell_B(i) \le 1$ for all i.*

5.2 Pricing the coverage constraint

Applying [BKNS22] to $\hat{I}$ is legal by Theorem 19(i) and returns some B with $\hat{p} \in \{0,1\}^n$ and total at most $n-1$. But it is free to place two copies of one type in one bundle, in which case that chore is absent from at least two bundles, has at least two owners, and the image under Φ^{-1} is not a partition. By Corollary 20 coverage is the sole residual difficulty, so it suffices to make duplicates unattractive.

Two ways of doing so suggest themselves. One may *stage* the allocation, dealing out the $n-1$ copies of a_1 together, then those of a_2, and so on, with each chore priced far above the last, so that an agent who has just taken a copy of the current chore is too well supplied to be selected again while copies of that chore remain; staging makes coverage automatic, since a stage placing its $n-1$ copies on distinct agents yields a coverage allocation by construction. Or one may simply charge for duplicates directly, adding a penalty proportional to the number of repeated copies a bundle holds. Neither works, and both fail for the same reason.

Example 21 (Duplicates are invisible to any price)**.** In the replica a second copy of a chore has marginal value exactly 0: if b has the type of a chore already represented in S then $\tau(S \cup \{b\}) = \tau(S)$, so $\hat{v}_i(S \cup \{b\}) = \hat{v}_i(S)$ for every agent i. The holder of a duplicate is therefore *indifferent* to it, not averse, which is the wrong sign: a selection rule is not repelled by the duplicate, it merely gains nothing from avoiding it.

Nor can a price supply the missing aversion. A charge levied on duplicates bites only through the arcs of the envy graph, and there it cancels: whatever an agent is charged for the duplicates she holds lowers every arc *out of* her by exactly what the charge raises the arcs *into* her from everyone else, so the two contributions annihilate around every cycle, and by Theorem 7(iii) no allocation is disqualified. A price can therefore work only if the agent who *holds* the duplicate does not feel it while somebody else does — and since prices are fixed before the allocation is chosen, while the allocation decides who holds the duplicate, requiring holder-blindness for every possible holder forces the price to zero.

Staging does not evade this. Its escalation separates *chores*, whereas the duplicate problem lives *within* one chore: the two copies at issue are copies of the same a_j, literally the same object, and no price assigned to chores can tell them apart. A further obstacle stands behind that one, and is fatal on its own — making one chore "much costlier" than another requires marginals outside $\{0, 1\}$, which leaves the dichotomous class and forfeits the very theorem the staging was meant to invoke.

5.3 What the three failures have in common

Each of the three techniques fixes a rule before the quantity it must respond to has been determined. Insertion fixes an owner before the consequences of that grouping are visible, and cannot later re-split a bundle. A selection rule fixes its criterion before knowing which allocation the criterion will pick out. A price fixes a charge before knowing who will hold the duplicate being charged for. In each case the rule is settled *ex ante* against a quantity determined *ex post*.

What survives is the requirement that a correct procedure decide late: it must be free to move already-allocated chores between bundles, and free to give up total cost while doing so. Enforcing coverage *structurally* rather than by price meets that requirement — if the copies of a chore simply cannot be placed on the same agent, coverage holds by construction and no charge is needed — and it is the direction the constructions in progress take.

6 What is Proved

Conjecture 2 is open. Five restrictions of it are not, and they are collected here because together they delimit what a general proof still has to reach. The first four are cases of the valuation class or of the instance; the fifth is a criterion, and it is the only one that says anything about three agents in general.

6.1 Four solved cases

Theorem 22 (Binary additive costs)**.** *Conjecture 2 holds when every c_i is additive with all singleton values in $\{0, 1\}$, that is $c_i(S) = |S \cap D_i|$ for some $D_i \subseteq M$.*

This is [LMS26]; the contribution here is only the observation that the class sits inside the dichotomous one.

Theorem 23 (Identical costs)**.** *If $c_1 = \cdots = c_n = c$ then every allocation is envy-freeable, and greedily adding each chore to a currently cheapest bundle yields $p^* \in \{0,1\}^n$.*

Proof For any permutation σ, $\sum_i c(A_{\sigma(i)}) = \sum_i c(A_i)$, so condition (ii) of Theorem 7 holds with equality and every allocation is envy-freeable. Moreover $w_{\mathcal{A}}(i,j) = c(A_i) - c(A_j)$ telescopes exactly along a path, giving $\ell_{\mathcal{A}}(u) = c(A_u) - \min_j c(A_j)$, so it suffices to keep $\max_i c(A_i) - \min_i c(A_i) \le 1$. That is an invariant of the greedy rule: if all costs lie in $\{\mu, \mu+1\}$ and the next chore joins a bundle attaining μ, its cost becomes μ or $\mu+1$ because marginals lie in $\{0,1\}$, and no other bundle moves. Observation 9 finishes. □

The case $n = 2$ also holds, with total subsidy at most 1; there the theorem is [BKNS22, Theorem 4] and ours is only the reduction, which consumes the identity $M \setminus A_j = A_i$ and so says nothing for $n \ge 3$.

Theorem 24 (Small bundles)**.** *Let $\mathcal{A}$ be envy-freeable with $c_v(A_i) \le 1$ for all $v, i \in N$. Then $p^* \in \{0,1\}^n$ and the total subsidy is at most $n-1$.*

Proof By Theorem 15, $\ell_{\mathcal{A}}(u) \le \max(0, \max_{v \ne u}[c_v(A_u) - c_v(A_v)]) \le \max(0, 1-0) = 1$. Observation 9 gives $p^* \in \{0,1\}^n$, and the endpoint of a heaviest path has $\ell = 0$. □

Corollary 25 (Fewer chores than agents)**.** *If $m \le n$ then Conjecture 2 holds, and a witness is computable by one minimum-cost bipartite matching.*

Proof Among allocations giving each agent at most one chore take one minimising $\sum_i c_i(A_i)$. Every permutation of its own bundles is an allocation of the same form, so Theorem 7(ii) makes it envy-freeable; and $|A_i| \le 1$ forces $c_v(A_i) \le 1$, so Theorem 24 applies. The bound is tight by Example 1, which has $m = n-1$. □

Definition 26 (Uniformly balanced family)**.** *A partition of M into n bundles is* uniformly balanced *if every agent values all n bundles within one unit of each other.*

Theorem 27 (Instances admitting a uniformly balanced family)**.** *Conjecture 2 holds for every instance admitting a uniformly balanced family, and a witness is obtained by assigning the bundles by one maximum-weight matching.*

Proof Assign by a maximum-weight matching; Theorem 7 makes the allocation envy-freeable, and uniform balance gives $w_{\mathcal{A}}(i,j) \ge -1$ for *any* assignment of that family, so Corollary 16 applies. □

This last case is much the widest of the four. It contains, for instance, every *symmetric* instance — one in which each $c_i(S)$ depends only on $|S|$ — since a partition into bundles of near-equal size shows agent i only the two values $f_i(q)$ and $f_i(q+1)$, which differ by a single marginal. It is nevertheless not everything: there are instances admitting no uniformly balanced family at all, and by Section 4 no criterion phrased on arcs alone can reach them.

6.2 A criterion at three agents

The fifth result is a criterion rather than a class, and it is the only statement in this report that bears on three agents in general. For a family $B = (B_1, \dots, B_n)$ — a partition of M into n possibly empty bundles, not yet assigned to agents — write

$$\mathrm{sp}_i(B) := \max_t c_i(B_t) - \min_t c_i(B_t), \qquad \Sigma(B) := \sum_{i \in N} \mathrm{sp}_i(B).$$

A family is uniformly balanced exactly when every $\mathrm{sp}_i(B) \le 1$.

Theorem 28 (Small total spread suffices)**.** *Let $n = 3$ and let σ be any minimum-cost assignment of a family B to the agents. If an arc of $G_{\mathcal{A}}$ has weight at least 2 then $\Sigma(B) \ge 4$; and if every arc has weight at most 1 but some two-path has weight 2, then $\Sigma(B) \ge 5$. In particular, if $\Sigma(B) \le 3$ then* every *minimum-cost assignment of B is good.*

Proof Normalise by $v_i(t) := c_i(B_t) - \min_s c_i(B_s)$ and put $x_i := v_i(\sigma(i))$, so that $w_{\mathcal{A}}(i,k) = v_i(\sigma(i)) - v_i(\sigma(k))$ and a minimum-cost assignment is one minimising $F(\sigma) = \sum_i x_i$. If $x_i \ge 1$ then v_i does not vanish at $\sigma(i)$, so its zero lies at some $\sigma(k)$ and the heaviest arc out of i is exactly x_i; if $x_i = 0$ every arc out of i is at most 0.

Suppose some arc has weight at least 2, so some $x_i \ge 2$, and let k hold the zero of v_i. Transposing i and k changes F by $v_k(\sigma(i)) - x_i - x_k$, non-negative by optimality, so $v_k(\sigma(i)) \ge x_i + x_k \ge 2$ and $\mathrm{sp}_k \ge 2$; with $\mathrm{sp}_i \ge x_i \ge 2$ this gives $\Sigma \ge 4$.

Suppose instead every arc is at most 1 and some two-path $i \to j \to k$ has weight 2. Then $x_i = x_j = 1$, $v_i(\sigma(j)) = 0$ and $v_j(\sigma(k)) = 0$, so $F(\sigma) = 2 + x_k$. Transposing i and j costs $v_j(\sigma(i)) + x_k$, forcing $v_j(\sigma(i)) \ge 2$ and $\mathrm{sp}_j \ge 2$. The three-cycle $i \mapsto \sigma(j)$, $j \mapsto \sigma(k)$, $k \mapsto \sigma(i)$ costs $v_k(\sigma(i))$, forcing $v_k(\sigma(i)) \ge 2 + x_k \ge 2$ and $\mathrm{sp}_k \ge 2$. With $\mathrm{sp}_i \ge 1$ this gives $\Sigma \ge 5$.

For the last claim, a minimum-cost assignment is envy-freeable by Theorem 7, so by Theorem 29 goodness is exactly the absence of an arc of weight 2 and of a two-path of weight 2, both of which force $\Sigma \ge 4$. □

Proposition 29 (Goodness at three agents)**.** *Let $n = 3$ and let $\mathcal{A}$ be envy-freeable. Then $\mathcal{A}$ is good if and only if every arc has weight at most 1 and every directed two-path has weight at most 1.*

Proof With no positive cycle, $\ell_{\mathcal{A}}(i)$ is the heaviest weight of a *simple* path leaving i; on three vertices such a path has length 0, 1 or 2. □

Theorem 28 strictly contains Theorem 27 at $n = 3$: a uniformly balanced family has $\Sigma \le 3$, but so do families with spread profiles such as $(2, 1, 0)$ and $(3, 0, 0)$, which are not uniformly balanced. It therefore reaches instances admitting no uniformly balanced family at all — the ones Section 4 shows no arc-based criterion can certify.

Remark 30 (What it leaves open)**.** Theorem 28 is conditional on a family with $\Sigma \le 3$ existing. Whether one always does is open, and is the single statement to which Conjecture 2 at $n = 3$ has been reduced. Two facts bound the question. If no uniformly balanced family exists then, since each agent can individually be brought to spread at most 1 by the greedy rule of Theorem 23, a family with $\Sigma \le 3$ must give some agent spread exactly 0. And for binary additive costs such an agent exists only when 3 divides $|D_i|$, so the question turns on a divisibility condition on the underlying sets. The full development is deferred.

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